LAB 4

Training and Deploying a Megalopa Detection Model on Raspberry Pi 5

Fork-based GitHub workflow | Google Colab | YOLO26n | USB webcam

Session	Duration	Focus	Marks
Session 1	2 hours	Fork, dataset, training, evaluation, ONNX export, Git push	60
Session 2	2 hours	Clone fork on Pi 5, USB-webcam inference, deployment evaluation	40
Total	4 hours	Complete cloud-to-edge workflow	100

Official student submissions: SecXX_GroupXX_Lab4_Session1.pdf and SecXX_GroupXX_Lab4_Session2.pdf

1. Laboratory objective

Students complete a reproducible object-detection workflow from a labelled megalopa dataset to live edge inference on Raspberry Pi 5. The instructor maintains a public upstream repository, while every student group works in its own fork.

- Inspect and validate a YOLO bounding-box dataset.
- Create train, validation, and test subsets.
- Train a lightweight YOLO26n detector in Google Colab.
- Interpret precision, recall, mAP, confusion matrices, and confidence curves.
- Export the trained PyTorch model to ONNX.
- Push models and selected results to the group fork.
- Clone the same fork on Raspberry Pi 5.
- Run static and live inference using a USB webcam and local display.

2. GitHub architecture

The laboratory uses the following repository relationship:

WUStuLab upstream → group fork → Colab training/push → Pi 5 clone

2.1 Instructor upstream

- Repository: WUStuLab/lab4-megalopa.
- Contains official dataset, validation material, notebooks, and Pi scripts.
- Students do not require Write access to this repository.
- The upstream main branch remains instructor-controlled.

2.2 Group fork

1. Every student must have a GitHub account.
2. One designated group member creates the group fork.
3. The fork owner adds the remaining group members as collaborators.
4. Session 1 clones and pushes to this fork.
5. Session 2 clones the same fork on Raspberry Pi 5.

3. Material supplied through Google Classroom/GitHub

Material	Purpose
Lab4_Instruction.pdf	Complete two-session laboratory instruction
Session 1 Colab notebook	Training, evaluation, export, answers, and PDF evidence
Session 2 Colab notebook	Pi walkthrough, evidence uploads, answers, and PDF evidence
YOLO box-label dataset	Shared training data
Validation images and video	Generalization and video testing
USB-webcam Pi script	Static and live inference on Raspberry Pi 5
Environment checker	Verify OpenCV, Ultralytics, display, and webcam

4. Session 1 - Dataset, training, evaluation, and fork push

Time	Activity
0-15 min	Student information, fork verification, and Colab setup
15-35 min	Dataset inspection and random five-image preview
35-50 min	Dataset split and training-parameter explanation
50-85 min	YOLO26n training
85-105 min	Metrics, plots, and model export
105-115 min	Image and video validation
115-120 min	Commit and push student_work/ to the group fork

4.1 Required Session 1 workflow

6. Enter group members, section, group number, fork owner, fork repository name, GitHub username, and commit email.
7. Install only the required Colab packages and verify the GPU.
8. Clone the public group fork without authentication.
9. Inspect dataset counts, image sizes, label pairing, coordinate ranges, and class IDs.
10. Preview five random original/annotated image pairs; rerunning the cell displays another sample.
11. Create a 70/20/10 split in temporary Colab storage.
12. Train YOLO26n using the instructor-defined parameters.
13. Display numerical metrics and all available YOLO plots.
14. Save best.pt, export ONNX, and verify both formats on the same image.
15. Validate on five test images, supplied validation images, and the supplied video.
16. Stage only student_work/, authenticate using Colab Secrets, and push to the group fork.
17. Print the completed notebook as SecXX_GroupXX_Lab4_Session1.pdf.

4.2 GitHub authentication for the final push

Cloning a public fork is anonymous. Pushing requires authentication. The GitHub account used for the push must own the fork or be a collaborator.

18. Create a fine-grained personal access token.
19. Limit repository access to the group fork.
20. Set Contents permission to Read and write.
21. Store the token in Colab Secrets as GITHUB_TOKEN.
22. Never paste the token into a visible notebook cell or PDF.

5. Session 2 - Raspberry Pi 5 deployment

The development kit already contains Raspberry Pi OS, a local screen, Python, Git, OpenCV, and Ultralytics. The camera is a USB webcam. Students do not prepare an SD card and do not use SSH.

Time	Activity
0-15 min	Verify Pi 5 system and USB webcam
15-30 min	Clone the group fork and verify the Session 1 commit/model
30-50 min	Verify software environment and run static inference
50-90 min	Run live USB-webcam inference and capture evidence
90-110 min	Compare .pt and ONNX; observe confidence and temperature
110-120 min	Complete Session 2 notebook and PDF submission

5.1 Required Session 2 workflow

23. Enter group/fork information and the Session 1 commit ID in the Session 2 notebook.
24. Record Raspberry Pi model, architecture, Python version, and initial temperature.
25. Verify the USB webcam and save a raw test image.
26. Clone the public group fork on the Pi.
27. Confirm the Pi commit matches the Session 1 commit ID.
28. Verify best.pt and ONNX files under student_work/models/.
29. Run static-image inference before live inference.
30. Run the local USB-webcam detection window; press S to save and Q to stop.
31. Compare .pt and ONNX under the same conditions.
32. Upload screenshots/photographs to the Session 2 notebook.
33. Print the completed notebook as SecXX_GroupXX_Lab4_Session2.pdf.

6. Assessment

Deliverable	Questions	Marks
Session 1 completed Colab PDF	27 embedded questions	60
Session 2 completed Colab PDF	20 embedded questions	40
Total	47 embedded questions	100

6.1 Evidence policy

Questions are marked together with the relevant output. An answer unsupported by the required code output, metric plot, screenshot, photograph, or result table may receive reduced credit.

7. Instructor preparation checklist

- Insert the final dataset and validation files.
- Verify every image-label pair and the intended class mapping.
- Test the full Session 1 notebook in a temporary student fork.
- Test .pt and ONNX loading on the actual Pi 5 kits.
- Test the USB webcam index, focus, lighting, and local GUI window.
- Confirm each group creates a fork before the session begins.
- Freeze the upstream repository before students fork it.
- Keep backup trained models and one backup Pi kit available.

8. Submission checklist

- Session 1 PDF contains all 27 answers, visible outputs, fork URL, and final commit ID.
- Session 2 PDF contains all 20 answers, uploaded Pi evidence, fork URL, and matching commit ID.
- No token, password, or private credential appears in either PDF.
- The model files are visible in the group fork under student_work/models/.

9. Reference links

- **GitHub: Fork a repository:** https://docs.github.com/en/pull-requests/how-tos/work-with-forks/fork-a-repo
- **GitHub: Personal access tokens:** https://docs.github.com/en/authentication/keeping-your-account-and-data-secure/managing-your-personal-access-tokens
- **Ultralytics: Model export:** https://docs.ultralytics.com/modes/export/
- **Ultralytics: Raspberry Pi guide:** https://docs.ultralytics.com/guides/raspberry-pi/